INTERFACE REQUIREMENTS SPECIFICATION

Software-Defined Radio

Document Control

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1. Introduction

1.1 Purpose

This document defines the interface requirements for the Software Defined Radio (SDR) system.

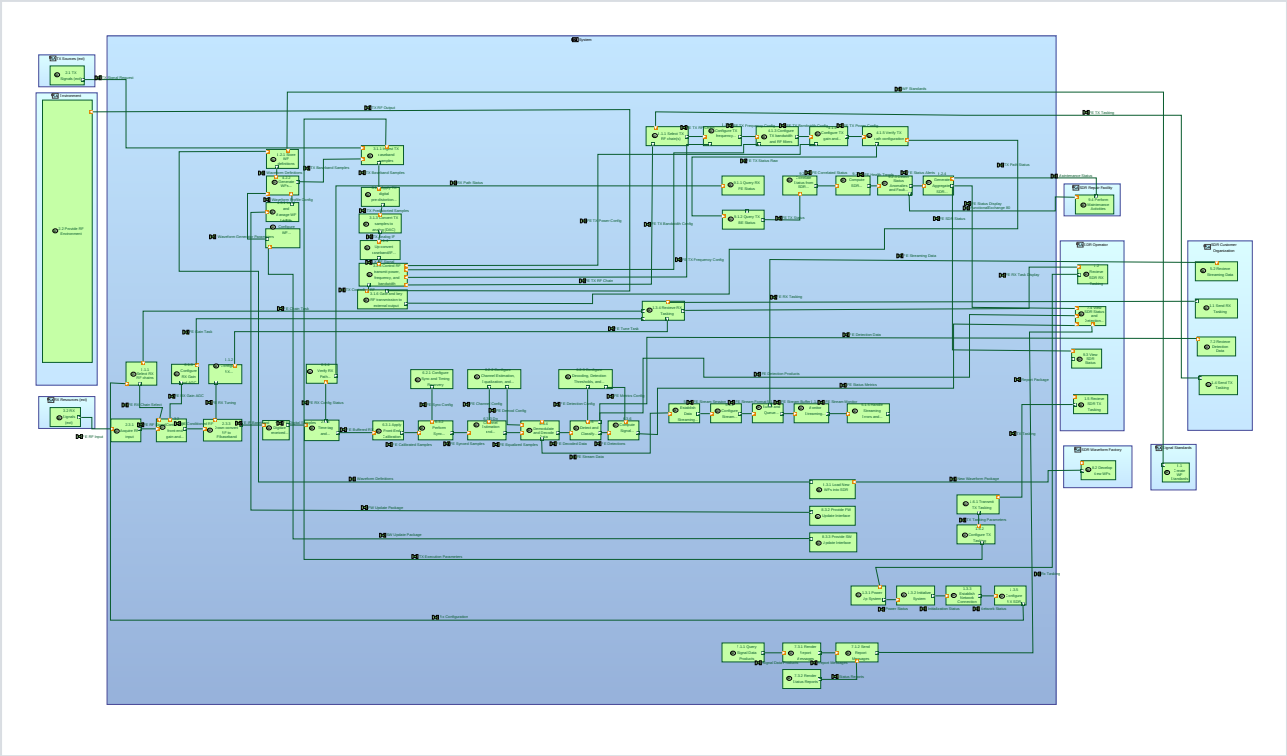
1.2 Scope

The requirements specified herein are derived from system architecture models and define external functional exchanges across the SDR system boundary.

2. System Architecture Diagram

The System Architecture Blank (SAB) diagram shown below provides the architectural context from which the external functional exchanges were derived. These exchanges form the basis of the interface requirements defined in this document.

Figure 1 — [SAB] Structure



3. Interface Identification

Interface ID	Source System	Destination System	Interface Type
SDR-External-Interfaces	External Actors	SDR System	Functional Exchange

Boundary Description: Defines interfaces that cross the SDR system boundary.

4. Interface Requirements

4.1 External Functional Exchanges

Requirement ID	Requirement Statement
REQ-SA-EXT-FE-001	The system shall receive RF Input from 3.2 RX Signals (ext).
REQ-SA-EXT-FE-002	The system shall receive Streaming Data from 5.1.3 Buffer and Queue Streaming Data.
REQ-SA-EXT-FE-003	The system shall receive RX Tasking from 1.1 Send RX Tasking.
REQ-SA-EXT-FE-004	The system shall receive RX Task Display from 1.3.4 Recieve RX Tasking.
REQ-SA-EXT-FE-005	The system shall receive Detection Data from 6.3.5 Detect and Classify Signals.
REQ-SA-EXT-FE-006	The system shall receive Detection Products from 6.3.5 Detect and Classify Signals.
REQ-SA-EXT-FE-007	The system shall receive Status Metrics from 6.3.6 Compute Signal Quality Metrics.
REQ-SA-EXT-FE-008	The system shall receive SDR Status from 9.2.4 Generate Aggregated SDR Status Views.
REQ-SA-EXT-FE-009	The system shall receive Status Display from 9.2.4 Generate Aggregated SDR Status Views.
REQ-SA-EXT-FE-010	The system shall receive TX Tasking from 1.4 Send TX Tasking.
REQ-SA-EXT-FE-011	The system shall receive TX Signal Request from 2.1 TX Signals (ext).
REQ-SA-EXT-FE-012	The system shall receive TX RF Output from 3.1.6 Gate and key RF transmission to external output.

Requirement ID	Requirement Statement
REQ-SA-EXT-FE-013	The system shall receive New Waveform Package from 8.2 Develop New WFs.
REQ-SA-EXT-FE-014	The system shall receive TX Tasking from 1.5 Recieve SDR TX Tasking.
REQ-SA-EXT-FE-015	The system shall receive Rx Tasking from 1.2 Recieve SDR RX Tasking.
REQ-SA-EXT-FE-016	The system shall receive Report Package from 7.1.2 Send Report Messages.
REQ-SA-EXT-FE-017	The system shall receive WF Standards from 8.1 Create WF Standards.
REQ-SA-EXT-FE-018	The system shall receive maintenance data from 9.2.3 Detect Status Anomalies and Fault Conditions.
REQ-SA-EXT-FE-019	The system shall receive Maintenance Status from 9.4 Perform Maintenance Activities.

5. Requirements Verification Traceability Matrix

Requirement ID	Requirement Statement	Verification Method	Verification Artifact	Success Criteria	Status	Notes
REQ-SA-EXT-FE-001	The system shall receive RF Input from 3.2 RX Signals (ext).	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-002	The system shall receive Streaming Data from 5.1.3 Buffer and Queue Streaming Data.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-003	The system shall receive RX Tasking from 1.1 Send RX Tasking.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-004	The system shall receive RX Task Display from 1.3.4 Recieve RX Tasking.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-005	The system shall receive Detection Data from 6.3.5 Detect and Classify Signals.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-006	The system shall receive Detection Products from 6.3.5 Detect and Classify Signals.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-007	The system shall receive Status Metrics from 6.3.6 Compute Signal Quality Metrics.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-008	The system shall receive SDR Status from 9.2.4 Generate Aggregated SDR Status Views.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-009	The system shall receive Status Display from 9.2.4 Generate Aggregated SDR Status Views.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-010	The system shall receive TX Tasking from 1.4 Send TX Tasking.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-011	The system shall receive TX Signal Request from 2.1 TX Signals (ext).	Test	TBD	TBD	Planned	

Requirement ID	Requirement Statement	Verification Method	Verification Artifact	Success Criteria	Status	Notes
REQ-SA-EXT-FE-012	The system shall receive TX RF Output from 3.1.6 Gate and key RF transmission to external output.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-013	The system shall receive New Waveform Package from 8.2 Develop New WFs.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-014	The system shall receive TX Tasking from 1.5 Recieve SDR TX Tasking.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-015	The system shall receive Rx Tasking from 1.2 Recieve SDR RX Tasking.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-016	The system shall receive Report Package from 7.1.2 Send Report Messages.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-017	The system shall receive WF Standards from 8.1 Create WF Standards.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-018	The system shall receive maintenance data from 9.2.3 Detect Status Anomalies and Fault Conditions.	Test	TBD	TBD	Planned	
REQ-SA-EXT-FE-019	The system shall receive Maintenance Status from 9.4 Perform Maintenance Activities.	Test	TBD	TBD	Planned	

6. Notes and Constraints

1. **TBD:** TBD

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Systems Engineering Lead	Program Manager
_____	_____
Date: _____	Date: _____